

ROLE OF NIGELLA SATIVA IN OPIOID DEPENDENCE TREATMENT

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ABSTRACT

Opioid dependence is one of the major social and psychiatric problem of society. Unfortunately there is no non-opiate treatment available. This study was conducted to find out the role of Nigella Sativa as a potential non-opiate treatment for opiate withdrawal. The study was conducted at the Department of Pharmacology, University of Karachi and Drug Rehabilitation Centre, R.H.C., Murad Memon Goth, Karachi. The study was based on DSM IV criteria for opioid dependence. Nigella Sativa 250 mg is found effective in long-term management of opioid dependence.

Keywords: Drug dependence, non-opiate, Nigella Sativa.

INTRODUCTION

Drug dependence is a condition in which an individual feels compelled to repeatedly administer a psychoactive drug repeated drug use appears to be a learned behaviour that is reinforced both by the pleasureable effects of the drug and by the negative effects of drug abstinence (withdrawal). These effects are the basis of drug craving in drug-dependent individuals. psychologic dependence is due to the positive reinforcement of drug use that results from the activation of neurons located in the nucleus accumbens. Physical dependence is a state in which continued drug use is required to prevent an unpleasant withdrawal syndrome hence physical dependence leads to negative reinforcement of drug use both psychologic and physical dependence appears to result from neuronal adaptation to the presence of the drug in different areas of the brain.

Increase in population, modernisation of society and increase in competition among various classes of society has increased the social and psychological problems, those people who can not cope with this menace usually fall prey to DRUGS to avoid bitterness of life. Opioids are the commonest of all the

drugs of abuse. Unfortunately no non opiate treatment is presently available for opioid dependence the drugs which are used presently are having their own side effects including a mild withdrawal syndrome.

This study was conducted to find a non opiate drug, which should not be having any euphoric or withdrawal effects and that can treat the opioid dependence syndrome effectively.

Drug addiction (drug dependence) is a serious health problem, in addition to the huge direct health cost (psychiatric and physical), there are massive costs in terms of crime, loss of earnings and productivity, and social damages. Reducing the extent of drug dependence is one of the major goals of medicine (Nutt, 1996).

Physical dependence and tolerance are hallmarks of opioid type of drugs. It is characterized by abstinence syndrome when opioid intake is abruptly terminated or when opioid antagonist is being used. Nowadays anti-psychotics (chlorpromazine and thioridazine) and anti-depressants (amitriptylline) are commonly used for controlling opioid abstinence syndrome in different hospitals at

Karachi but their efficacy is scant. There is as yet no truly safe and effective treatment for the opioid dependent patients to manage withdrawal and craving, and certainly none that has universal applicability (Shulman *et al.*, 1998).

Because no single medication is appropriate for every individual for treating their opioid addiction, it is important that clinicians have a variety of the therapeutic agents available to them. Rational medication therapy begins with an understanding of not only the disease state generally, but also the specific dynamics of addiction process that affect the overall success of treatment.

Opioid detoxification/maintenance therapy remains the primary pharmacological approach for the treatment of opioid dependence. At present only two medications, methadone and naltrexone are currently approved for the treatment of opioid addiction. However, detoxification from methadone maintenance is slow and frequently accompanied by minor abstinence syndrome. Also opiate agonists have their own potential for abuse and withdrawal symptoms. Naltrexone is a competitive antagonist at the mu-opioid receptors and is used to maintain abstinence in detoxified opioid addicts. It has found limited acceptance by population targeted for treatment, and its successful utilization generally requires highly motivated patients. Hence it necessitates use of a non-opiate treatment for opioid detoxification and dependence.

As *Nigella Sativa* has been found to have calcium channel blocking ability, inhibiting action potential in neuronal as well as peripheral tissues and calcium channel blockers have been proved to be effective in opioid withdrawal syndrome so purpose of present study was to evaluate the effects of *Nigella Sativa* in opioid dependence.

MATERIALS AND METHODS

This study was carried out at the Drug Rehabilitation Centre, RHC Murad Memon

Goth, Malir, Karachi and in the Department of Pharmacology, University of Karachi.

A total of 35 male opiate addicts who were seeking treatment for opioid dependence were consecutively admitted between September 2001 to September 2003. All were admitted for 12 days to treat acute opiate withdrawal syndrome and then treated for opioid dependence as outpatients for 12 weeks.

Selection of Patients

All patients in the study were selected according to the following criteria:

Inclusion Criteria

1. Males between 21 and 45 years of age seeking treatment for opioid dependence for a duration of three months.
2. Following routine clinical criteria indicating opioid addiction were observed:
 - a. Self reported duration of opioid dependence of at least four months.
 - b. An average of two or more episodes of opioid use per day.
 - c. Physical evidence of recent intravenous drug use (tracks).
 - d. Urine toxicology positive for opiates at entry to the study.
 - e. A rating of two or greater on a self-reported level of withdrawal scale 12 hours after the last opioid use.
3. Ability and willingness to give informed consent.

Exclusion Criteria

1. Psychiatric illness e.g., anxiety, neurosis, phobia, obsessive, compulsive neurosis, and hysteria.
2. Self reported current dependence on alcohol or other major drug of abuse like sedatives, hypnotics (including benzodiazepines), cocaine, or amphetamines.
3. Acute liver or cardiovascular diseases.
4. Current enrolment in an opiate treatment programme.
5. Any debilitating disease.

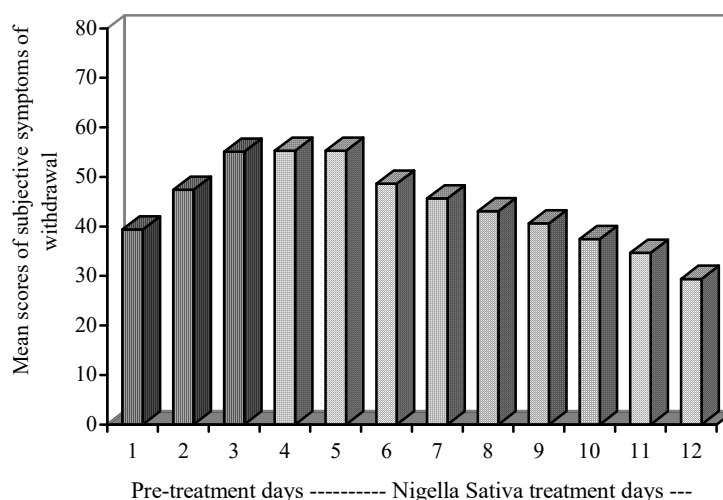


Fig. 1: Effects of Nigella Sativa 250 mg Treatment on Subjective Symptoms of Withdrawal from Opioids in Group-I patients during 12 days Stay in Hospital.

Effects of Nigella Sativa treatment on the mean $\pm$ SEM scores of 38 symptoms of withdrawal of the 12 days sample in 25 opioid addicts at days after the third in patient day of withdrawal. Students' t-test comparing change in symptoms from pre-treatment inpatient day-3 to treatment day-12 indicated trend decrease in symptoms to be reported on day-12 of admission.

Materials

1. Dried black seeds of Nigella Sativa (Kalonji) were purchased from Majeed Brothers, Lajpat Road, Hyderabad, and were cleaned off from adulterant materials and were ground with an electric grinder into coarse powder.
2. Empty capsules manufactured locally were purchased from open market.
3. The frontline opiates test strips obtained from Boehringer Mannheim Pakistan (Pvt.) Ltd., Ch-B/Lot No. 28739531 and expiry in February 2004.

Treatment Schedule

The selected patients with opioid dependence were kept on 250 mg Nigella Sativa orally TID.

The patients received single blind placebo capsule (orally) containing ferrous sulphate powder of same colour, size and shape for the drug, during day-1 and day-2 of admission.

They were observed and rated for the presence and absence of opioid withdrawal signs and symptoms experienced during the previous 24 hours by an observer.

Thereafter each treatment, group received single blind capsule, containing 250 mg Nigella Sativa on day-3 of admission (treatment day-1). After the initial administration, patients with a positive response to the drug in terms of opiate withdrawal signs and symptoms and without producing side effects were given the drug upto day-12 of admission (treatment day-10). Diazepam 5 mg was prescribed for some patients having anxiety. The ratings covered the withdrawal signs and symptoms occurring in the previous 24 hours. All the patients received their respective treatment upto day-12 of admission during their stay in hospital. After that patients were discharged on the same treatment and advised to attend OPD weekly upto twelve weeks.

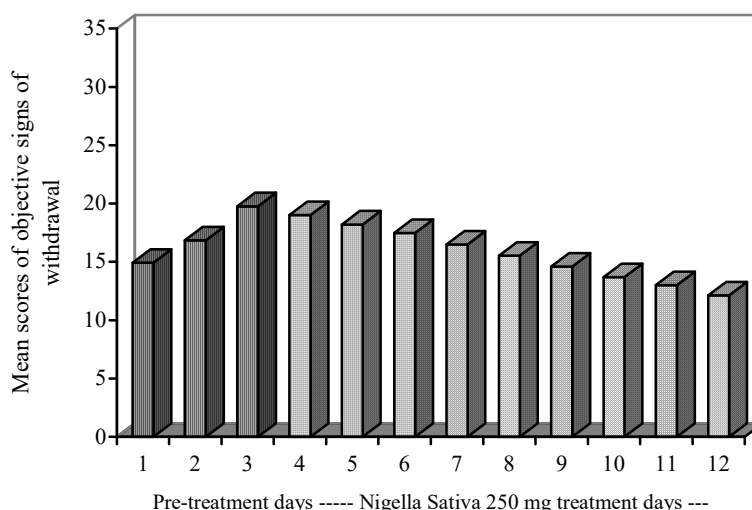


Fig. 2: Effects of *Nigella Sativa* 250 mg Treatment on Objective Signs of Withdrawal from Opioids in patients during 12 days Stay in Hospital.

Effects of *Nigella Sativa* treatment on the mean $\pm$ SEM scores of 18 signs of withdrawal of the 12 days sample in 25 opioid addicts at days after the third inpatient day of withdrawal. Students' t-test comparing change in signs from pre-treatment inpatient day-3 to treatment day-12 indicated trend decrease in symptoms to be reported on day-12 of admission.

Method

Measures

The selected patients were enrolled, data and progress of the patients were recorded, which included the parameters for abstinence as well as protracted withdrawal for opiate dependence.

Subject-Reported Measures

It was in the form of modified subjective opiate withdrawal scale (MSOWS), which contained 38 opiate withdrawal symptoms. Subjects indicated the degree to which they had experienced each symptom during the past 24 hours on a five point scale in which 0=not at all, 1=a little, 2=moderately, 3=quite a bit, and 4=extremely (maximum possible total score was 152). The ratings for individual item were summed for a total score each scale.

Observer Rated Measures

It was in the form of objective opiate withdrawal scale (OOWS,) containing 18

observable physical signs. An independent observer observed and rated the presence and intensity of signs on a five point grade scale in which 0=not at all, 1=a little, 2=moderately, 3=quite a bit, and 4= extremely (Hiltrunen *et al.*, 1995).

PHYSIOLOGICAL PARAMETERS

It includes the pulse rate, systolic blood-pressure, diastolic blood pressure, temperature, respiratory rate, body weight, and caloric intake (Martin and Jasinski, 1969).

Urine Analysis Measures

Urine samples were collected on days-1 and 12 of admission and then on weeks-4, 8 and 12. All samples were collected under staff observation to deter bogus urine samples and tested immediately for opioids by using one-step dip and read chromatographic test strips (Frontline opiates test strips).

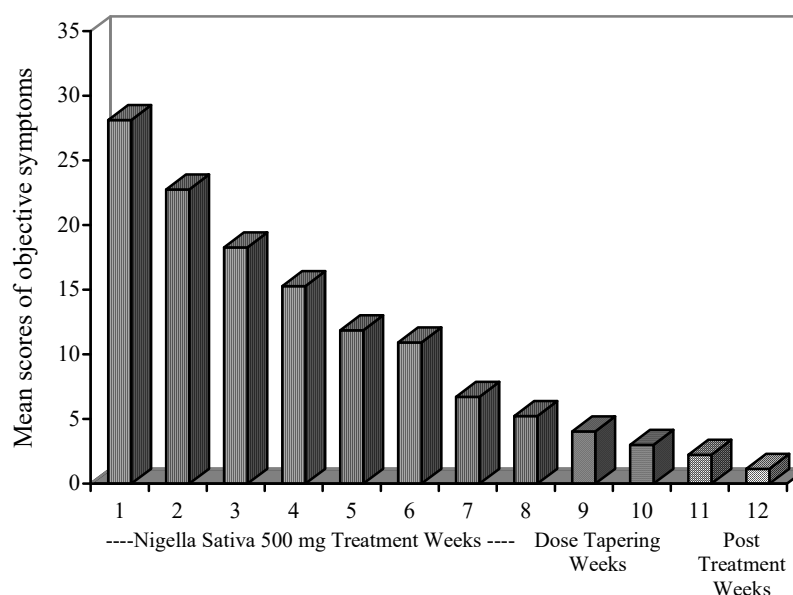


Fig. 3: Effects of Nigella Sativa 250 mg Treatment on Subjective Symptoms of Protracted Abstinence in patients with Opioid Dependence during 12 weeks of Follow up.

Effects of Nigella Sativa 250 mg treatment on the mean $\pm$ SEM scores of subjective symptoms of protracted abstinence in 25 patients after every week during 12 weeks of follow up. Students' t-test comparing change in symptoms from pre-treatment inpatient day-3 to week-12 indicated trend decrease in symptoms to be reported on week-12 of follow up.

OBSERVATIONS AND RESULTS

Total 35 patients were enrolled in the study; out of these, 10 (28.7%) patients were dropped during first and second weeks of treatment.

All patients were men ranging in age from 21-44 years ($x = 33.152 \pm 0.421$). They all expressed interest in discontinuing the use of opioid and gave written consents to join the study that required an abrupt withdrawal from opioids after admission to hospital. They had a mean of 4.957 ± 0.553 years history of opioid consumption (range 1-15 years). All had subjective symptoms and objective signs of opiate withdrawal and urine specimens showing positive results when tested with frontline opiates dipstick-subjects of Nigella Sativa 250mg treatment showed minor adverse effects.

Our final analysis applies to 35 patients who completed the protocol: Nigella Sativa 250 mg orally T.I.D.

Some patients received Diazepam 5 mg for night-time sedation.

The results show a cumulative score of opiate withdrawal signs and symptoms on day-1 to day-12 of admission and then on week-I to week-12 as outpatient. The degree of withdrawal signs and symptoms were assessed according to the scoring system described in methodology.

Twenty-five patients were given a placebo treatment on day-1 and day-2 of admission. Thereafter from day-3 to day-12 of admission the patients received 250 mg Nigella Sativa orally, three times daily.

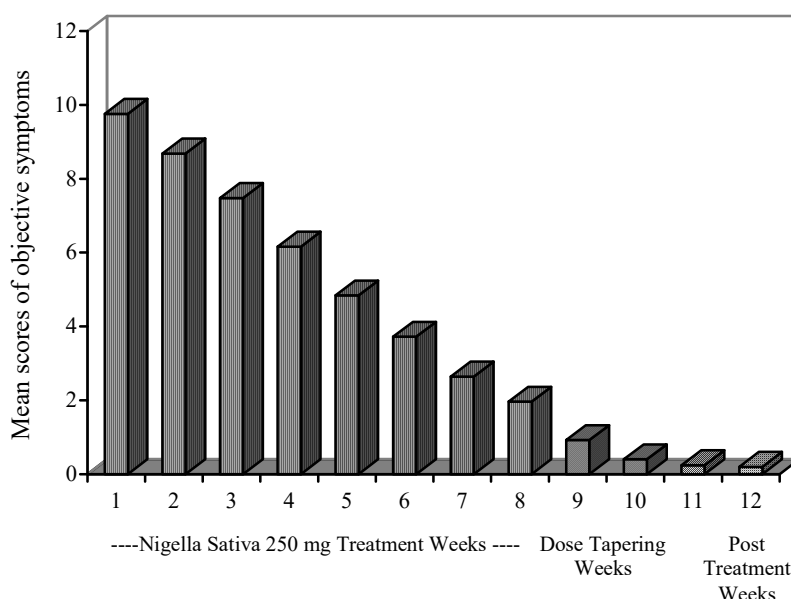


Fig. 4: Effects of *Nigella Sativa* 250 mg Treatment on Objective Signs of Protracted Abstinence in patients with Opioid Dependence during 12 weeks of Follow up.

Effects of *Nigella Sativa* 250 mg treatment on the mean $\pm$ SEM scores of objective signs of protracted abstinence in 25 opioid addicts after every week of 12 weeks of follow up. Students' t-test comparing change in signs from pre-treatment inpatient day-3 to week-12 indicated trend decrease in signs to be reported on week-12 of follow up.

Placebo had no significant effect on cumulative scores of symptoms of acute withdrawal from opioids. A mean score of 39.4 ± 7.48 was obtained on day-1, which had reached to a peak of 55.08 ± 12.65 on day-3 of admission. However admission of *Nigella Sativa* 250 mg produced a decrease in opiate withdrawal symptoms from 55.08 ± 12.65 on day-3 to 29.4 ± 5.02 on day-12 of admission.

Similarly placebo had no significant effects on the cumulative score of signs of acute withdrawal from opioids. A mean score of 14.92 ± 3.35 was obtained on day-1, which reached a peak of 19.76 ± 4.58 on day-3 of admission. However, admission of *Nigella Sativa* 250 mg produced a rapid decrease in opiate withdrawal signs from 19.76 ± 4.58 on day-3 to 12.12 ± 2.60 on day-12 of admission.

The physiological parameters including

pulse rate, systolic blood pressure, diastolic blood pressure, temperature and respiratory rate were decreased significantly from day-3 to day-12 of admission. There were also increase in body weight and caloric intake from day-3 to day-12 of admission, while the pupillary diameter was increased from 3.28 ± 0.35 on day-3 to 2.70 ± 0.32 on day-12 of admission.

Urine toxicology was also decreased significantly from day-3 to day-12 of admission.

Nigella Sativa 250 mg treatment regimen used in this study shows a significant effect on withdrawal from opioids as substantiated with clinical condition, physical appearance, and the cumulative scores of withdrawal signs and symptoms daily.

The patient was discharged on the same

treatment after 12 days of admission and advised to attend OPD weekly for the follow up.

The cumulative scores of symptoms of protracted abstinence was decreased from 55.08 ± 12.65 on day-3 to 1.16 ± 1.84 after 12 weeks of follow up. Similarly, the cumulative score of signs of protracted abstinence was decreased from 19.76 ± 4.58 to 0.20 ± 0.64 after 12 weeks of follow up.

The physiological parameters including pulse rate, temperature, respiratory rate decreased and papillary diameter were increased from day-3 to week-12 of follow up. There was a decrease in systolic and diastolic blood pressure from day-3 to week-12 of follow up. The caloric intake was increased from 1.41 ± 0.41 to 2.70 ± 0.13 after 12 weeks of follow-up reported on week-12 of follow up.

DISCUSSION

Heroin dependents are those who continue to use heroin in the face of difficulties, they knew or believe to be caused by its use such as health, legal and inter-personal difficulties. They typically use heroin daily, develop tolerance to its effects, and experience withdrawal symptoms on abrupt cessation of use. About one quarter of people who have ever used heroin developed dependence.

Dependence does not end when the drug is removed from the body (detoxification) or when the acute post-drug taking illness dissipates (withdrawal). Rather, the underlying addictive disorder persists, and this persistence produces a tendency to relapse to active drug taking.

Methadone maintenance treatment (MMT) is the most extensively researched form of maintenance treatment for opioid dependence. The effectiveness of treatment for opioid dependence would ideally be assessed through randomized controlled trials (RCTs). Only five such trials have ever taken place in

the 35 years since MMT was introduced. All five trials involved small number of patients who were rarely followed for longer than one year. The effectiveness of MMT in observational studies of community treatment programmes has not been as impressive as that in the RCTs, indicating that methadone treatment in routine use is not a panacea for opioid dependence. About half of those who enter treatment left within 12 months, and some of those continued to one heroin and other illicit drugs though much less frequently than before they entered treatment (Hall *et al.*, 1998).

Maintenance with pure opioid antagonists such as naltrexone has been shown to be effective in opioid dependent people for whom failure to comply with treatment has major personal consequences (e.g., opioid dependent health professionals) but pure opioid antagonists have not proven popular with the wider population of opioid dependent people, in whom low rates of compliance have been a major difficulty. Naltrexone has almost no agonist effects and will not satisfy craving or relieve protracted withdrawal symptoms. For these reasons, naltrexone treatment does not appeal to the average heroin addict especially those with less motivation to remain opioid free (O'Brien and McLellan, 1998).

This study was the first project launched in the Department of Pharmacology and Therapeutics, University of Karachi. The objective for this project was to search non-opiate treatment of opioid dependence for long-term management; a treatment which would be more safe, less hazardous and more acceptable to opioid addicts.

Prior to this few studies were conducted at Basic Medical Sciences Institute, J.P.M.C., Karachi, in which role of calcium channel blockers was observed in the treatment of opioid dependence syndrome. These studies were carried out by Baloch (1991), Mahesar (1994), Salat (1998), and Ansari (1999) in which effect of calcium channel blocker in treatment of opioid dependence was evaluated

in animal and patients.

Baloch (1991) and Mahesar (1994) observed the effect of verapamil and felodipine in morphine dependent animals subjected to naloxone *in vivo* and *in vitro*. They observed that calcium channel blockers were effective in reducing the abstinence *in vivo* and *in vitro* effects. These observations led to pilot project on morphine addict patients.

First clinical study was conducted by Salat in 1998, who compared the effects of calcium channel blocker, verapamil with thioridazine and amitriptyline, contemporary treatment prevalent in Karachi in the management of acute opioid abstinence syndrome.

He observed that verapamil showed a highly significant improvement in signs and symptoms of abstinence. The patients, who were admitted during the study with previous history of opioid abstinence without treatment, expressed that they did not experience these withdrawal effects. He concluded that verapamil therapy is safe, effective, and more pronounced in treating the acute opioid withdrawal syndrome than amitriptyline and thioridazine.

Second clinical study was done by Ansari (1999) in which the effects of verapamil and clonidine compared with thioridazine and chlorpromazine in opioid abstinence syndrome were observed. He observed that the effects of verapamil and clonidine to decrease the signs and symptoms of acute withdrawal from opioids were highly significant when compared with chlorpromazine and thioridazine.

Hence, in the light of previous clinical studies, this single blind study was proposed to observe the effects of *Nigella Sativa* in long-term management of opioid dependence. Each patient received treatment initially for 12 days during his stay in hospital. Then each patient was advised for weekly follow up as out patient and the same treatment was continued

till eight weeks then the dose of drug was gradually tapered off during next two weeks period after that patient was followed up further for two weeks without giving any drug.

A placebo for the drug was given on day-1 and 2 of admission, only to observe and confirm the opiate withdrawal syndrome on first three days of admission. *Nigella Sativa* was administered in dose of 250 mg and 500 mg three times daily, in the two groups. Each group comprised of 25 patients. *Nigella Sativa* produced a decrease in opiate withdrawal signs and symptoms.

Nigella Sativa 250 mg reduced the subjective symptoms from pretreatment day-3 scoring rate of 55.08 ± 12.65 to 29.4 ± 5.02 at day-12. Similarly, objective signs were also reduced from pretreatment day-3 scoring rate of 19.76 ± 4.58 to 12.12 ± 2.60 at day-12.

Regarding the physiological parameters decrease was observed in pulse rate, systolic blood pressure, diastolic blood pressure, and temperature, but all were within normal physiological ranges. Respiratory rate and pupil diameter also decreased. Both body weight and caloric intake increased.

Similarly, *Nigella Sativa* 500 mg was given to 25 patients three times daily throughout the treatment period after abrupt discontinuation of opioid administration. It also produced rapid decrease in opiate withdrawal signs and symptoms. The subjective symptoms were reduced from pretreatment day-3 scoring rate of 63.2 ± 13.57 to 14.56 ± 8.13 at day-12. Similarly objective signs were also reduced from pretreatment day-3 scoring rate of 25.52 ± 3.08 to 7.72 ± 2.35 at day-10.

There were 20 symptoms in the opiate withdrawal questionnaire used in previous studies with minimum score of zero to maximum 80. Grading of intensity of symptoms was from 0-4 with increasing severity. While the state used in our study had 38 subjective symptoms with a maximum

score of 152. The additional symptoms were added for the more comprehensive assessment of the state of acute withdrawal as well as for assessing the state of protracted withdrawal after the acute abstinence. The additional symptoms were such as decrease in appetite, alertness, cheerfulness, calmness, patience, relaxation, and clear thinking, disorientation, carefreeness, drug craving, dysphoric mood, feelings of anxiety, increased sensitivity to pain, low psychomotor speed, pounding heart, sadness and shooting up.

Similarly, there were only six signs in opiate withdrawal questionnaire in previous with from 0-4 with minimum score of zero to maximum 24. Grading of intensity of signs were increasing severity. While the scale used in our study had 18 objective signs with a maximum score of 72. The additional signs were added for the more comprehensive assessment of the state of acute withdrawal as well as for assessing the state of protracted withdrawal after the acute abstinence. The additional signs were such as air, hunger, anorexia, insomnia, mydriasis, and tremor.

As it was proved in an in vitro study that *Nigella Sativa* is having calcium channel blocking effect, along with analgesic, spasmolytic, anti-microbial, and anti-diarrhoeal etc., effects so this drug was used in this study.

CONCLUSION

It is concluded that this drug is effective in long term management of opioid dependence.

It not merely cures the opioid dependence but also cures the infections and weakness from which majority of addicts suffer. It is suggested that further long-term follow up studies are needed to evaluate the benefit of this drug in maintaining the patients opioid free. In addition, various biochemical and physiological evidences are required to further strengthen the effectiveness of this non-opiate

drug in long-term management of opioid dependence. Keeping in view the mode of action of opioids, pharmacological effects of *Nigella Sativa* and results of present study, it is concluded that *Nigella Sativa* is an effective treatment of opioid dependence it is cheap and readily available and far more less in cost than other drugs presently available for the treatment of opioid withdrawal syndrome its anti-allergic, anti-bacterial, spasmolytic and anti-nociceptive effects and nutritious rich amino acids are more than of a benefit to an opioid addict, who usually requires these remedies and nutritious supplements. It is advised to conduct more research on this valuable seed in relation to its pharmacological potential.

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